

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: Departmental Digital Resource & Knowledge Hub

Version: 1.0

Prepared For: Department-Level Academic Deployment

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for a centralized web-based academic resource and notice management system for a department.

The system will:

- Centralize academic materials
 - Provide structured upload and approval workflows
 - Prevent misinformation via official notice board
 - Support bulk onboarding via Excel
 - Enforce role-based access control
-

1.2 Scope

The system is a web-based portal that:

- Enables three user roles:
 - Admin (HOD)
 - Lecturer
 - Student
- Provides structured digital repository
- Implements approval moderation for student uploads
- Supports advanced filtering and search
- Includes a digital notice board
- Supports Excel-based bulk onboarding

The system is intended for internal departmental use.

1.3 Definitions

Term	Definition
USN	University Seat Number
RBAC	Role-Based Access Control
Pending	Awaiting approval
Approved	Visible to users
Rejected	Denied publication

2. Overall Description

2.1 Product Perspective

This is a standalone web application built using:

- Frontend: Web-based interface
 - Backend: REST API (e.g., FastAPI)
 - Database: Relational DB (e.g., MySQL)
 - File Storage: Server-based structured storage
-

2.2 User Classes

1. Admin (HOD)

Full system authority.

2. Lecturer

Content publisher and moderator.

3. Student

Content consumer and contributor (pending approval).

2.3 Operating Environment

- Web browser (Chrome, Edge, Firefox)
 - Server deployment (local or cloud)
 - Relational DB backend
-

3. System Features

3.1 Authentication & Authorization Module

Features:

- Secure login for all roles
 - Role-based dashboard redirection
 - Password hashing (bcrypt)
 - JWT-based session control
 - Logout functionality
-

3.2 User Management Module

Supports:

- Student self-registration
 - Admin manual user creation
 - Admin Excel-based bulk onboarding
 - Duplicate USN prevention
 - Unique Employee ID enforcement
-

3.3 Smart Repository Module

Organized by:

- Semester
- Subject Code
- Unit

Supported Formats (Strictly Enforced):

- PDF
 - PPT / PPTX
 - DOC / DOCX
 - JPG / PNG
-

3.4 Gatekeeper Approval Module

- Student uploads → Status: Pending
 - Lecturer/Admin → Approve or Reject
 - Only Approved → Visible in search
-

3.5 Advanced Search & Filtering

Filters:

- Semester
- Subject Code
- Professor
- Resource Type

Only Approved resources appear.

3.6 Digital Notice Board

- Post official notices
 - Highlight for 24 hours
 - Display on homepage dashboard
-

4. Functional Requirements

4.1 Admin Functional Pages & Capabilities

4.1.1 Admin Login Page

- Secure login
-

4.1.2 Admin Dashboard

Displays:

- Total Students
 - Total Lecturers
 - Total Resources
 - Pending Upload Count
 - Recent Notices
-

4.1.3 User Management Page

Functions:

- View all users

- Search users
 - Edit user details
 - Deactivate account
 - Delete account
-

4.1.4 Bulk Upload Page (Excel)

Functions:

- Upload .xlsx file
 - Validate schema
 - Validate duplicates
 - Insert valid records
 - Display summary:
 - Total rows
 - Inserted
 - Failed
 - Duplicates
-

4.1.5 Manual User Creation Page

Admin can:

- Add Student
 - Add Lecturer
 - Assign role
-

4.1.6 Review Pending Uploads Page

Admin can:

- View file
 - Download file
 - Approve
 - Reject
-

4.1.7 Notice Management Page

Admin can:

- Create notice
 - Edit notice
 - Delete notice
 - Set priority
 - Set expiry
-

4.1.8 Resource Management Page

Admin can:

- View all resources

- Delete inappropriate resources
 - Filter by status
-

4.2 Lecturer Functional Pages & Capabilities

4.2.1 Lecturer Login Page

Secure login.

4.2.2 Lecturer Dashboard

Displays:

- Own uploaded resources
 - Pending approvals
 - Recent notices
-

4.2.3 Upload Resource Page

Fields:

- Title
- Semester
- Subject Code
- Unit
- File Upload

Behavior:

- Auto status = Approved
-

4.2.4 Review Pending Uploads Page

Lecturer can:

- View pending student uploads
 - Approve
 - Reject
-

4.2.5 Notice Posting Page

Lecturer can:

- Create notice
 - Edit own notices
 - Delete own notices
-

4.2.6 My Resources Page

Lecturer can:

- View uploaded files
 - Delete own files
-

4.3 Student Functional Pages & Capabilities

4.3.1 Student Registration Page

Fields:

- USN
- Name
- Semester
- Password

Validations:

- Unique USN
 - Proper format
-

4.3.2 Student Login Page

Secure login.

4.3.3 Student Dashboard

Displays:

- Search bar
 - New notices
 - Recently added resources
 - Upload button
-

4.3.4 Search & Filter Page

Students can:

- Filter by semester
 - Filter by subject
 - Filter by professor
 - Download approved files
-

4.3.5 Upload Notes Page

Fields:

- Title
- Semester
- Subject
- Unit
- File Upload

Behavior:

- Status = Pending
 - Not visible publicly
-

4.3.6 My Uploads Page

Students can:

- View uploaded files
 - See status (Pending / Approved / Rejected)
 - Delete own pending uploads
-

5. Non-Functional Requirements

5.1 Security

- Password hashing (bcrypt)
 - JWT session tokens
 - Role-based access control
 - SQL injection prevention (ORM)
 - File MIME validation
 - Upload size limits
 - Secure Excel parsing
 - Activity logging
-

5.2 Performance

- Search results under 2 seconds
 - Support up to 1000 concurrent users
 - Indexed DB queries
-

5.3 Scalability

- Modular architecture
 - Storage abstraction
 - Pagination support
-

5.4 Availability

- 99% uptime target (production)
 - Backup strategy recommended
-

5.5 Usability

- Clean dashboard UI
 - Responsive design
 - Dropdown-based filtering
-

6. External Interface Requirements

6.1 User Interface

- Web browser based
- Role-based dashboards

- Clean grid view for files
-

6.2 Hardware Interface

- Standard web server
 - File storage disk
-

6.3 Software Interface

- Database connection (MySQL)
 - File handling service
-

7. Data & Storage Requirements

7.1 User Data

- USN (unique)
 - Employee ID (unique)
 - Name
 - Role
 - Semester
 - Password hash
-

7.2 Resource Data

- Title
 - Semester
 - Subject
 - Unit
 - File path
 - Status
 - Uploaded by
 - Timestamp
-

7.3 Notice Data

- Title
 - Description
 - Created by
 - Created date
 - Priority
-

8. Architecture Overview

Layered Architecture:

1. Presentation Layer (Frontend)
2. API Layer

3. Business Logic Layer
4. Database Layer
5. File Storage Layer

RBAC middleware controls access.

9. Security, Privacy & Compliance Requirements

- Protect student personal data
 - Do not expose passwords
 - Encrypt sensitive data
 - Prevent unauthorized file access
 - Enforce approval before publication
 - Audit log critical actions
-

10. Deployment & Operational Requirements

MVP Deployment:

- Local server
- Single instance backend
- Local storage

Production Expansion:

- Cloud hosting
 - Cloud storage (S3)
 - SSL certificate
 - Automated backups
 - Load balancing
-

11. Future Enhancements

- Email notifications
 - Download analytics
 - Popular resources section
 - Mobile PWA support
 - Semester rollover automation
 - Advanced reporting dashboard
 - File virus scanning
-

12. Appendix

Supported File Types

Extension	Allowed
-----------	---------

PDF	Yes
-----	-----

PPT/PPTX	Yes
----------	-----

DOC/DOCX	Yes
----------	-----

Extension Allowed

JPG/PNG Yes

Others No

Final System Summary

This system delivers:

- Centralized academic repository
 - Structured approval workflow
 - Controlled student contributions
 - Bulk onboarding capability
 - Role-based governance
 - Advanced search and filtering
 - Official notice management
-